

Research and Working Notes

Student Motivation & Engagement Survey, Week 3

SafeX Solutions Internship, Group 43
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These are the working notes kept during the project. They are informal by design and record what was considered, tried, and set aside, alongside the decisions that shaped the final deliverables.

Framing the Problem

The Week 3 brief called for a survey on “student motivation and learning habits” feeding recommendations for an educational platform. Two framings were considered:

- **Diagnostic:** measure current motivation levels across students. Simpler to design, less actionable for platform design.
- **Design-driven:** identify which specific factors affect motivation, so the platform can act on them. Harder to design, but more useful to Group 43’s brief.

Chose the design-driven framing. The survey should not just ask “how motivated are you” but “what would make you more motivated” and “what breaks your motivation.”

Question Design Decisions

Considered several question formats before settling on the final 15.

Kept

- Likert-scale items on motivation, engagement, feedback, group work, routines. Fast for respondents, easy to aggregate.
- Open-ended prompts on “biggest motivation killer,” “one thing to change,” and “support that would help.” Slower to code but where the most useful signal usually comes from.
- A multi-select on engagement factors so respondents could name more than one improvement.

Dropped or set aside

- An initial idea to ask for GPA or attendance percentage was dropped. It would have made responses less anonymous and would have discouraged completion.
- A ranking question for platform features was set aside because it takes longer to complete than a multi-select and produced similar signal in a quick pilot.
- A question on preferred class timings was cut for scope, since it does not map to any platform feature.

Design goals for each question

- Neutral wording. Not “do you agree that class is boring” but “I feel motivated to attend my classes.”
- One idea per question. Compound questions were split (for example, motivation and interest were separated into Q2 and Q3).
- Ordered from easy to harder. Demographics and simple Likert items came first; open-ended prompts came later.

Distribution Notes

The survey was distributed through:

- University WhatsApp groups (main channel).
- Class chats within COMSATS.
- Personal network of student friends and peers.

Kept the distribution informal and short. The message that got the best response rate was a two-line request with a direct link, sent in the evening rather than the morning. Follow-up reminders were sent once, forty-eight hours after the initial share, to any group where response count had stalled.

Sources Consulted (Background)

Consulted the following to inform question phrasing and framing (background reading, not cited in the report):

- Ryan and Deci's Self-Determination Theory framing (autonomy, competence, relatedness) to shape the feedback and relevance items.
- Fact-check methodology pages of AFP, AP, and Reuters were reviewed in the same week for a separate PNP task and reinforced the value of showing the reasoning behind each conclusion. That habit carried into the analysis workbook.
- Public survey design guides on Google Forms best practice (question order, avoiding double-barrelled items).

Analysis Decisions

Agreement percentage definition

Agreement % = (Agree + Strongly Agree) / Total responses. This is the simplest possible summary and works for all Likert items in the survey. A weighted mean (assigning 1 to 5 to the scale points) was considered but not used, because it hides where the mass sits and turns a bimodal distribution into a misleading middle value.

Thematic coding of open-ended answers

Read all responses to each open-ended question, listed every distinct point, then merged points that were essentially the same. Kept the raw quotations alongside the themes in the workbook so any theme could be traced back to source responses without re-reading the entire form.

Handling the response drop-off

Q12 to Q15 have n=10, not n=16. Two options were considered:

- Drop the last four items from analysis.
- Report them separately with n=10 flagged clearly.

Kept them, since they contain the follow-through data and the platform feature suggestions, which are useful even at n=10. Flagged the smaller sample in the report and on the workbook rows.

Findings, Rough Cuts

Initial findings list before edits:

- Feedback is the strongest motivator identified.
- Relevance drives motivation, and its absence drains it.
- Consistency is a common gap.
- Students want more practical, less theoretical teaching.

- Emotional and interpersonal factors matter alongside academic ones.
- Distraction peaks mid-session.
- Interactive learning is the most requested change.

Collapsed to three primary findings (feedback, relevance, follow-through) and two secondary patterns (practice-over-theory, emotional layer) because the report needed a clear hierarchy. The distraction and interactive-learning points fit inside the follow-through and additional-patterns sections without needing their own top-level heading.

Recommendation Mapping

Each of the five design recommendations was checked against a specific data point before being included:

- **Feedback in the core loop** traces to Q5 (94% agreement) and Q11 (feedback and clear goals theme).
- **Relevance made visible** traces to Q4 (87.5% agreement) and the “real-world impact tracker” response in Q15.
- **Accountability scaffolding** traces to Q12 (10% follow-through) and Q15 (micro-goal tracking, streak building).
- **Interactive-first content** traces to Q14 (50% each for interactive activities and smaller groups) and Q10 (theme: practice over theory).
- **Recognise the emotional layer** traces to Q9 (“degrades me”) and Q11 (“kindness,” “mental and emotional support”).

Rejected: a recommendation about “recommendation algorithms” was drafted, then cut. Nothing in the data pointed to it.

Loose Ends and Ideas for Later

- Rerun a shorter follow-up survey to test the “real-world impact tracker” concept with more respondents.
- If time permits, compare responses by year of study to see whether Year 1 and Year 3 students differ meaningfully in what motivates them. Sample size is too small to say confidently in this round.
- The emotional-layer finding sits uncomfortably next to the platform-design frame, since a platform cannot manufacture kindness. Worth thinking through how tone and language in notifications and instructor guidance can support this indirectly.

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